

Problem 1

- a. Let A_i be the event that you answer the i 'th question correctly. Then the overall number of questions you answer right is $Q = \sum_{i=1}^3 1_{A_i}$, and thus:

$$\mathbb{E}[Q] = \sum_{i=1}^3 \mathbb{E}[1_{A_i}] = \frac{3}{4} + \frac{1}{2} + \frac{1}{4} = 1.5.$$

- b. By independence of 1_{A_i} 's, the variance will be:

$$\text{Var}(Q) = \sum_{i=1}^3 \text{Var}(1_{A_i}) = \frac{3}{4} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{4} \times \frac{3}{4} = \frac{5}{8}.$$

- c. The problem hereby asks us to calculate $\mathbb{P}(A_3|Q = 2)$. We can use the definition of conditional probability here:

$$\begin{aligned} \mathbb{P}(A_3|Q = 2) &= \frac{\mathbb{P}(Q = 2 \cap A_3)}{\mathbb{P}(Q = 2)} \\ &= \frac{\mathbb{P}(A_1 \cap A_2^c \cap A_3) + \mathbb{P}(A_1^c \cap A_2 \cap A_3)}{\mathbb{P}(Q = 2)} \end{aligned}$$

Note that the event $\{Q = 2\} = \{A_1 \cap A_2 \cap A_3^c\} \cup \{A_1 \cap A_2^c \cap A_3\} \cup \{A_1^c \cap A_2 \cap A_3\}$, and since 1_{A_i} 's are independent we get that:

$$\begin{aligned} \mathbb{P}(Q = 2) &= \frac{3}{4} \times \frac{1}{2} \times \frac{3}{4} + \frac{3}{4} \times \frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{2} \times \frac{1}{4} = \frac{13}{32} \Rightarrow \\ \mathbb{P}(A_3|Q = 2) &= \frac{\frac{3}{4} \times \frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{2} \times \frac{1}{4}}{\frac{13}{32}} = \frac{4}{13}. \end{aligned}$$

Grading Comments:

- Part b: Many students tried to use the formula $V(X) = E(X^2) - (E(X))^2$ to solve for this problem. As $E(X)$ is solved in part a, $(E(X))^2$ is straightforward. However, solving for $E(X^2)$ is not so because since $X = 0, 1, 2, 3$, we need to find $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, $P(X = 3)$, which needs a bit of calculation work. For example, $P(X = 1)$ includes three possibilities of realization: first question is correct and the last two questions are wrong, the second question is correct and other two questions are wrong, and the third question is correct while the first two questions are wrong. You need to add up probabilities of these three possible outcomes to have $P(X = 1)$. Same for other cases. Comparatively, the approach given by the solution is much easier.
- Part c: A common mistake is fail to calculate the joint probability of getting two questions correct while getting the third question correct. Note that if you get two questions correct

and at the same time the third question is correct, then we must have that either first or second question is correct. Another common mistake is to omit the correctness of the third question when calculating the conditional probability. Note that to calculate $P(Q = 2)$ you have to take the possibility that the first two questions are correct and the third question is wrong into consideration, while in the numerator $P(Q = 2 \cap A_3)$ you don't (since the third question must be correct).

Problem 2

- a. Let A_i be the event that the outcome of the i -th throw is not 6. Then the event that no 6's appear in the first k throws corresponds to the event $\cap_{i=1}^k A_i$, and therefore we get:

$$\mathbb{P}(\text{No head in first } k) = \prod_{i=1}^k \mathbb{P}(A_i) = \left(\frac{5}{6}\right)^k.$$

- b. The number of throws until we see a 6 for the first time N is distributed geometrically with parameter $\frac{1}{6}$, thus

$$\mathbb{E}[N] = \frac{1}{1/6} = 6.$$

- c. This part can be exactly rephrased into one of our homework / precept problems! Think of having two contestants, first wins if 1 comes up whereas the second wins when 6 comes up. This question is the same as asking what is the probability that second contestant wins. As we have already examined, there is a myriad of ways of answering this question. Let us denote the event the first 6 comes up before the first 1 or 2 as A . Also let us denote the first time that either 1, 2 or 6 come up as T , and the value of the outcome on the k -th throw by a random variable Y_k . Then, we can first calculate the probability of getting a 6 before 1 or 2 conditioned on the event $T = k$:

$$\mathbb{P}(A|T = k) = \frac{\mathbb{P}(Y_1 \neq 1, 2, 6, Y_2 \neq 1, 2, 6, \dots, Y_{k-1} \neq 1, 2, 6, Y_k = 6)}{\mathbb{P}(T = k)} = \frac{\left(\frac{1}{2}\right)^{k-1} \times \frac{1}{6}}{\mathbb{P}(T = k)}.$$

Having calculated this, to get $\mathbb{P}(A)$ all we need to do is to apply the law of total probability:

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(A|T = k) \mathbb{P}(T = k) = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^{k-1} \times \frac{1}{6} = \frac{1}{3}.$$

Alternative solution:

Let us also show a solution using first step analysis. Let us denote the event that you throw 6 before the first 1 as A and the number that comes up on i -th throw by Y_i :

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A|Y_1 = 1, 2)\mathbb{P}(Y_1 = 1, 2) + \mathbb{P}(A|Y_1 = 6)\mathbb{P}(Y_1 = 6) + \mathbb{P}(A|Y_1 \neq 1, 2, 6)\mathbb{P}(Y_1 \neq 1, 2, 6) \\ &= 0 \times \frac{2}{6} + 1 \times \frac{1}{6} + \mathbb{P}(A) \times \frac{3}{6} \Rightarrow \\ \mathbb{P}(A) &= \frac{1}{3}.\end{aligned}$$

Grading Comments:

The most common mistakes were:

- part a: No justification of how to get the final answer. It's necessary to say the probability of each event and the independence.
- part b: No justification of how to get the final answer. You have to say it is a geometric distribution and then you can use the mean of it.
- part c: (1) It is a common mistake that students get the wrong conditional probability $\mathbb{P}(A|T = k)$. (2) Some students only calculated the conditional probability $\mathbb{P}(A|T = k)$ without summing them up.

Problem 3

Let us denote the craving time for ice cream by $T \sim \text{Exp}(1)$.

a. This corresponds to the event $\{T > 1\}$ and thus:

$$\mathbb{P}(T > 1) = e^{-1} \approx 0.37$$

b. The information given in this problem is that $\{T > 1\}$, thus the problem reduces to:

$$\begin{aligned}\mathbb{P}(T \leq 2|T > 1) &= 1 - \mathbb{P}(T > 2|T > 1) = 1 - \frac{\mathbb{P}(T > 2, T > 1)}{\mathbb{P}(T > 1)} \\ &= 1 - \frac{\mathbb{P}(T > 2)}{\mathbb{P}(T > 1)} = 1 - \frac{e^{-2}}{e^{-1}} = 1 - e^{-1} \approx 0.63.\end{aligned}$$

c. Let us denote the event that the craving comes on exactly during the first 6 minutes of the first hour as A_k , thus we have to calculate the probability of the event $\cup_{k=1}^{\infty} A_k$. Note that $A_k = \{T \in (k-1, k-0.9]\}$ and $A_i \cap A_j = \emptyset$ for $i \neq j$. Thus we get:

$$\begin{aligned}\mathbb{P}(\cup_{k=1}^{\infty} A_k) &= \sum_{k=1}^{\infty} \mathbb{P}(A_k) = \sum_{k=1}^{\infty} [\mathbb{P}(T > k-1) - \mathbb{P}(T > k-0.9)] = (e - e^{0.9}) \sum_{k=1}^{\infty} (e^{-k}) \\ &= (1 - e^{-0.1}) \sum_{k=0}^{\infty} (e^{-k}) = \frac{1 - e^{-0.1}}{1 - e^{-1}} \approx 0.15.\end{aligned}$$

Alternative Solution: One could also arrive at this answer using a different logic. We know that the craving has to arrive during some hour with probability 1, therefore all we have to calculate is that if the craving arrives during the k -th hour then it is in the first 6 minutes of it. Let's denote the event that the craving arrives during the k -th hour by H_k , then:

$$\begin{aligned}\mathbb{P}(\cup_{k=1}^{\infty} A_k) &= \sum_{k=0}^{\infty} \mathbb{P}(A_{k+1}|H_{k+1})\mathbb{P}(H_{k+1}) = \sum_{k=0}^{\infty} \mathbb{P}(T \leq k + 0.1 | k < T \leq k + 1)\mathbb{P}(H_{k+1}) \\ &= \sum_{k=0}^{\infty} \frac{\mathbb{P}(k < T \leq k + 0.1)}{\mathbb{P}(k < T \leq k + 1)}\mathbb{P}(H_{k+1}) = \frac{e^{-k} - e^{-k-0.1}}{e^{-k} - e^{-k-1}} \sum_{k=0}^{\infty} \mathbb{P}(H_{k+1}) = \frac{1 - e^{-0.1}}{1 - e^{-1}}.\end{aligned}$$

Note: $\mathbb{P}(T = a) = 0$ for all $a \geq 0$, thus one would get the exact same solution if all the strict inequalities were replaced with non-strict inequalities and vice versa. That is whether your interpretation of the statement was that the arrival of your craving within the first hour means $T \in (0, 1]$ or $T \in [0, 1)$, the answer should be the same.

Grading Comments:

- Part a: Many students only get the results for $\mathbb{P}(T \leq 1)$, although it seems trivial to get $\mathbb{P}(T > 1)$, you'd better write the solution the problem ask next time.
- Part b: A common mistake for this problem is fail to identify the problem is a conditional probability rather than a join probability.
- Part c: The most serious problem is that many students cannot find what is the event you want to work on and how to write it in a mathematical form.